
Large Language Models for Climate-Related Text Analysis: a review

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Abstract

This mini-paper analyzes the ClimateIE framework [7], which proposes a dataset and methodology for structured information extraction in climate science, including tasks such as entity recognition, relation extraction, and taxonomy alignment. We begin with a broader contextualization of LLMs in sustainability, followed by a review of the theoretical foundations underlying these models. We then discuss the core problems that ClimateIE aims to address, describe its methodology, and present its main results. Finally, we provide a critical analysis of its limitations and discuss possible directions for improvement.

1 Introduction

The accelerated growth of sustainability as a field has highlighted the urgent need for more organized and consistent taxonomies. New terms emerge continuously, while existing concepts are often misused or interpreted inconsistently, sometimes enabling practices such as greenwashing. In response, initiatives such as the EU Taxonomy [2] aim to introduce structure and standardization to sustainability-related activities.

At the same time, a large share of climate and sustainability knowledge remains embedded in unstructured textual sources, including scientific publications, policy documents, and corporate reports. Transforming this information into structured and actionable knowledge remains a significant challenge, as traditional approaches such as manual analysis or simple text classification fail to capture complex relationships between variables, scenarios, and domains.

Recent advances in natural language processing (NLP), particularly large language models (LLMs), have enabled more sophisticated approaches to addressing these challenges. Previous research has explored the use of LLMs for climate-related tasks, including information extraction, question answering, and benchmarking [1, 3, 7]. Within this context, ClimateIE proposes a structured information extraction framework that aims to identify entities, model relationships, and align extracted knowledge with domain-specific taxonomies.

2 Problem Definition

The task addressed in this work is to extract structured information from unstructured climate-related documents. Specifically, given a corpus of raw textual data, the main goal is to identify

relevant entities, relationships, and attributes, and map them into a consistent format. This problem is challenging due to the specific vocabulary, variability in writing styles, and the implicit nature of many relationships in climate science texts.

3 Methodology

3.1 Theory

Large language model (LLM) pipelines rely on a sequence of transformations that convert raw text into structured and interpretable representations. This process typically begins with text pre-processing and tokenization, followed by representation learning through neural architectures and semantic alignment using embedding. Tokenization is a fundamental step in NLP pipelines, where raw text is decomposed into smaller units (tokens) that can be processed by a model. Formally, we have that a text sequence x is mapped into a sequence of tokens:

$$x \rightarrow (t_1, t_2, \dots, t_n)$$

In modern LLMs, subword tokenization methods such as Byte Pair Encoding (BPE) or Unigram Language Modeling (ULM) are widely used [5], as they balance vocabulary size and robustness to specific terms. This is important in scientific domains like climate science and sustainability, where specialized terminology frequently appears. After tokenization, tokens are transformed into vector representations through embedding layers. Each token t_i is mapped to a dense vector $e_i \in \mathbb{R}^d$:

$$e_i = \text{Embedding}(t_i)$$

These embeddings encode semantic and contextual information, allowing the model to capture the relationships between words and concepts. Transformer-based architectures leverage self-attention mechanisms to compute contextualized representations [5]. The core operation is given by:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V$$

Where Q (queries) represents the information being queried, K (keys) represents the information used for matching, and V (values) contains the information to be aggregated. Depending on the architecture, language models can be categorized into encoder-only, decoder-only, or encoder-decoder models. Decoder-only models, such as Llama-3.3 (used in the ClimateIE framework), model the probability of a sequence token autoregressively:

$$P(x) = \prod_{t=1}^n P(x_t \mid x_1, \dots, x_{t-1})$$

Decoder-only models are trained using a maximum likelihood, where the model learns to predict the next token in a sequence. This is commonly implemented using the cross-entropy loss, defined as:

$$\mathcal{L} = - \sum_{t=1}^n \log P(x_t \mid x_1, \dots, x_{t-1})$$

3.2 ClimateIE framework

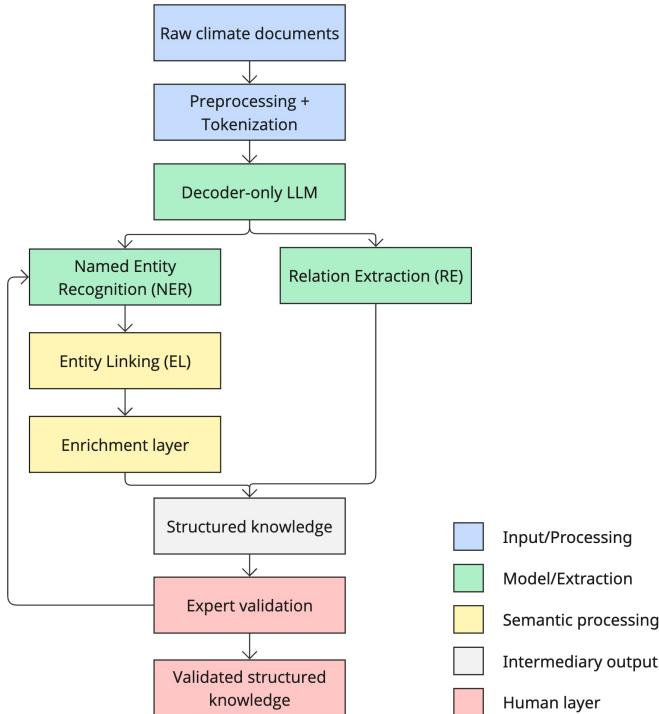


Figure 1: ClimateIE pipeline (adapted from [7])

The ClimateIE framework is a pipeline designed to extract structured information from climate science literature. It combines large language models with expert-guided validation to identify entities, relationships, and taxonomy. The framework focuses on three core information extraction tasks: Named Entity Recognition (NER), Relation Extraction (RE), and Entity Linking (EL). NER identifies domain-specific climate entities such as variables, models, locations, and datasets, while RE captures relationships between these entities. Finally, EL maps extracted entities to a standardized knowledge base, specifically the GCMD+ taxonomy [6], ensuring consistency and semantic normalization across extracted information.

The pipeline is supported by a hybrid annotation process that combines automatic extraction with human validation. A subset of 25 documents is annotated by domain experts, while a larger corpus of 625 publications is generated using AI-assisted methods, resulting in both human-annotated and AI-assisted datasets [4]. The final output consists of structured entities, relationships, and linked concepts, which can be used to support downstream applications such as climate informatics.

4 Results and Critical Analysis

The ClimateIE framework demonstrates strong empirical performance in extracting structured information from climate-related text across NER, RE and EL tasks. Among the evaluated models, Llama-3.3–70B achieves the best overall results, outperforming both proprietary models such as

GPT-4o and other open-source alternatives like DeepSeek-V3. In NER tasks, Llama-3.3 reaches a strict F1-score of approximately 0.378 and a relaxed F1-score of 0.501, surpassing DeepSeek-V3 (0.331 strict F1) and GPT-4o (0.291 strict F1). These results indicate that the model is particularly effective at identifying domain-specific entities under both exact and flexible matching criteria. Additionally, Llama-3.3 maintains competitive or superior performance across RE and EL tasks, achieving the highest F1 scores as well. This is especially interesting given that it outperforms larger proprietary systems, suggesting that domain alignment and task-specific design may be more important than model scale.

Despite these results, the framework presents some limitations. First, relying on a small manually annotated dataset (25 documents) may introduce bias and limit its ability to generalize. Expanding the expert annotated corpus could significantly improve performance. Second, sustainability and climate science often have semantic ambiguity. Terms like “renewable energy” and “low-carbon energy” are often used interchangeably despite referring to different concepts, while expressions like “carbon neutral” and “net zero” are sometimes used inconsistently, implying distinct mitigation strategies. These overlaps show how sustainability terminology is highly context-dependent, creating challenges for both human interpretation and automated systems, which may struggle to distinguish between subtly different meanings. Third, as highlighted in the paper, hallucination and factual inconsistency are still critical challenges in LLM-based systems, particularly in climate applications where precise terminology is essential. Fourth, the pipeline models NER, RE, and EL as sequential tasks, which may lead to error propagation. Mistakes in earlier steps can cascade and affect downstream predictions, reducing overall consistency.

5 Conclusion

This work reviewed the ClimateIE framework as an example of how LLMs can be applied to structure climate-related knowledge from unstructured text. The results show that open-source models such as Llama-3.3 can outperform larger proprietary systems in domain-specific tasks, reinforcing the importance of model-task alignment over scale alone. At the same time, key challenges remain, including data limitations, semantic ambiguity, hallucination risks, and error propagation across sequential pipelines. These findings suggest that advancing LLM applications in climate science requires richer and more structured datasets, clearer domain-specific taxonomies, and more integrated modeling approaches.

References

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